Gamified Learning Platform For Rural Education Software

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***Abstract*—The fast development of online learning platforms has provided the opportunities to enhance student learning by means of personalised and interactive learning systems. However, con traditional learning technologies tend to lose the attention of students, especially in resource-limited situations. This paper offers SHIKSHASETU, a web-based learning plat, which is gamified form meant to increase student motivation, engagement, and achievements in various subjects. The system proposed combines learning through games mentions like experience points(XP), leaderboards and levels interactive quizzes in a form of a guided academic system. The platform is based on a lightweight Flask architecture supports multi-role access for students and teachers, real-time performance monitoring, and multilingual content presentation ShikshaSetu uses a modu as opposed to traditional systems lar game engine which dynamically changes the format of questions challenges, encouraging involvement and engagement engagement. Offline-like accessibility is also supported by the system customization and localization, solving problems encountered in various learning situations Usability assessment and interaction metrics shows that there is a great progress in interaction with students and motivation. The results indicate that gamified learning systems have the ability to overcome the deficiencies in the conventional learning approach engage and create a more immersive and individualized learning experience. Some of the keywords are Gamification, Educational Technology, and so on A Learning System, Interactive Learning, Student Engagement, Web Application .**

I. Introduction

Digital technologies in education have integrated to bring about radically changed the conventional pedagogical practices. In spite of these innovations, student engagement is still a significant dilemma, especially science and technical subjects. Research has shown that traditional learning teaching methods not invariably keep students interested, leading to in decreased motivation and learning performance[6]. Gamification which refers to the integration of game design has opened up to become an exciting possibility to transfer elements into non-game contexts strategies to deal with this challenge. With the use of such mechanics as Gamified systems are intended to provide as rewards, levels, and challenges improve the learning outcomes and user engagement[7] A number of studies have shown the beneficial effect of gamification and student motivation and participation. For example, gamified learning settings have been demonstrated to raise involvement and academic achievements compared to the old-fashioned ways of teaching[10].However, many existing systems are not personalized, available, and real-time performance tracking.This paper aims to deal with these limitations offers ShikshaSetu, an interactive learning platform that is a game planned to offer a stimulating, dynamic, and inclusive educational experience. The system incorporates various within, multilingual assistance, real-time analytics, and subjects a unified platform

II. Literature Review

Gamified learning platforms development has been widely discussed in recent years, especially in the context of enhancing student interest and motivation. Traditional Although digital learning systems are effective in the delivery of content, are often not interactive and do not hold the attention of the learner over extended periods. Educational technology studies have shown to made it clear that game design elements like as points, levels, and leaderboards can significantly enhance user participation and learning outcomes by introducing ele ments of challenge and reward into the learning process[7]. Furthermore, the studies of gamification in teaching point out its capability to impact behavioral and cognitive areas of learning. Systems with organized challenges and feedback mechanisms have proved to be successful in improving student performance and engagement as compared to traditional teach ing methods[10].Nevertheless, actual implementations tend to be lacking a unified approach, with many platforms applying gamification elements that lack a well-defined design frame-

work, making them incon sistent outcomes. The other significant area is the development Score calculation of lightweight and easy to use learning management systems. The necessity of non-systems is highlighted in the latest research only social, yet also environmental friendly, including low-resource settings. Lightweight web-based archi modular designs and tectures have been demonstrated to enhance usability, minimize system complexity, and augment accessibility to a wider spectrum of users.Furthermore, multilingual support has grown very pertinent in the contemporary educational plat forms. It has been shown that presentation of content in different languages is beneficial languages increases inclusivity and allows learners of to better interact with people of different linguistic backgrounds. with digital systems. Nevertheless, there are numerous existing platforms that continue to have poor language adaptation and can only be used in a limited way diverse educational contexts. Despite the great advancements that have been achieved in these. Gamification, there is a lack of integration of gamification, Multilinguality, and real-time performance monitoring into one cohesive system.

III. SYSTEM ARCHITECTURE

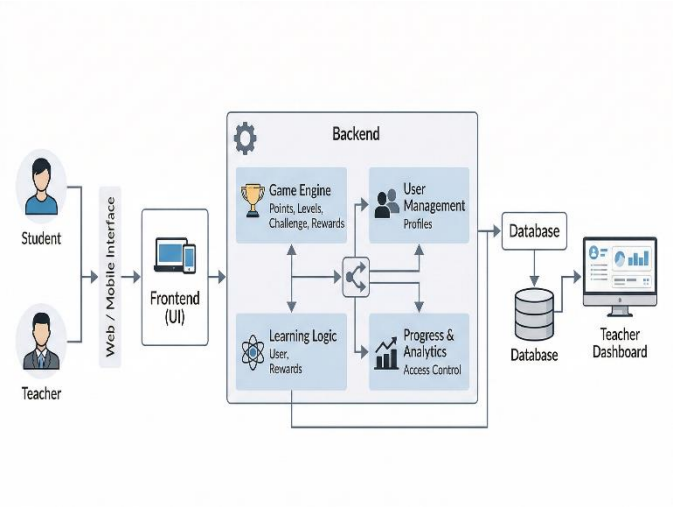


Fig. 1.*System Architecture Diagram*

The proposed system has a modular architecture with the following main components: frontend, backend, and game engine.

- **Backend System**

The backend part responsible for user authentication, session management and age data storage is implemented using Flask and SQLAlchemy. It provides role based access control for students and teachers. It provides secure and scalable operations.

- **Game Engine**

A dynamic game engine is the central part of the system and it handles:

- Real-time feedback
- Adaptive difficulty

The engine covers a wide range of subjects and grades for personalized learning. Frontend Interface

The user interface is built using HTML and CSS, and JavaScript offers:

- Interactive dashboards
- Leaderboards
- Notifications
- Multilingual translation features

IV. IMPLEMENTATION DETAILS

- ***Frontend Development***

ShikshaSetu user interface has been designed in a user-friendly manner to make it more interactive and accessible to provide a dynamic learning experience with the students. The frontend is built using popular web software such as HTML5, CSS3 and JavaScript. HTML5 gives the structural layout of the application, identifying features of dashboards, etc quiz interfaces, and navigation elements. Responsive layouts, color schemes and animations with CSS3 enhance visual appeal and provide gamification features like level indicators and progress bars. JavaScript plays a significant part in delivering dynamic interactions on the platform. It provides real time updates like score change, XP gain and leaderboard positionings without needing to reload the whole page. Additionally, asynchronous communication with the backend ensures a smooth user experience and gameplay. Multilingual support is implemented on the frontend level and provides language files in JSON. When the user clicks on a preferred language, everything in the interface is dynamically up dated, making accessibility easy to the users across linguistic backgrounds.

- ***Backend and Database***

The backend of the system is built with the Flask framework, which is a flexible and lightweight way to run operations on a server. Flask API routes are used to handle things like user authentication, managing sessions, and other game-related tasks like making quizzes and keeping track of scores. The database is a relational database that keeps track of users' progress in games and their performance. Users, subjects, game sessions, and scores are all important entities. The database is set up in a way that makes it easy to get and save data, which makes analytics and real-time possible. This architecture guarantees scalability and reliability, which means that the system can handle more than one user at a time without slowing down..

- ***Game Engine Integration***

The game engine is a key element of the system which controls quiz logic, scoring, and progres sion systems. Questions are dynamically generated by the engine according to the topics

chosen and level of difficulty. It evaluates user responses and scores by correctness and response time. Gamification features of experience points (XP), The engine calculates levels, and achievements. For example, users can gain XP by their performance, which has been used to enhance level development and leaderboard. This organized reward system promotes ongoing participation. and motivation. Learning is interactive and guaranteed by the game engine adaptive and gives instant user feedback after every interaction.

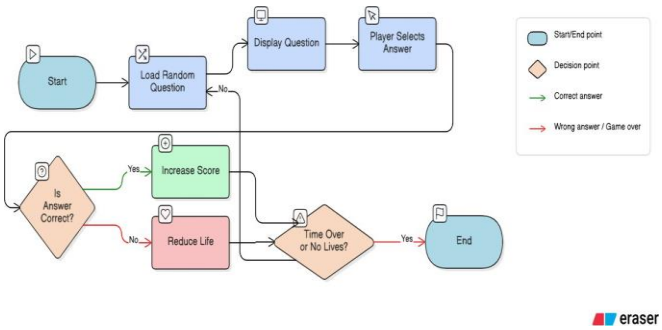


Fig. 2.*Game Engine Worlflow*

- ***Performance Tracking and Leaderboard***

The system has a built-in leader board that is ranked users depending on their performance measures like total score, XP, and accuracy. Such aspect brings in a competitive ele ment, motivating users to improve their performance. Continuous performance tracking is in place. user activity such as attempts at quizzes, accuracy prices, and session times. These indicators are presented on user dashboards, which would enable the students to monitor their progress time. The analytics modules and the leaderboard are created to real time update, so that the users get immediate feedback on their performance.

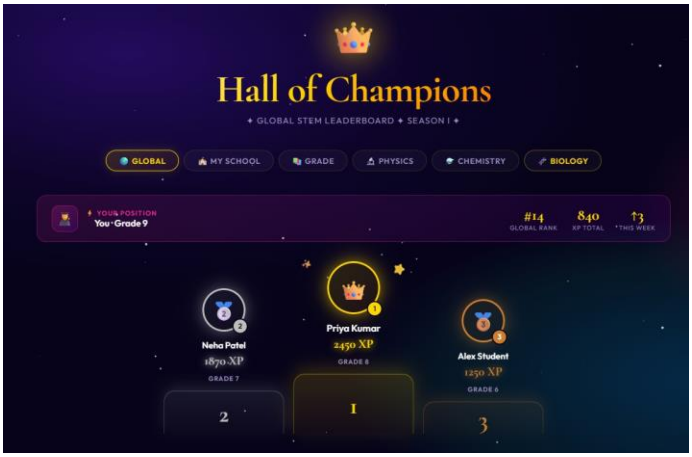


Fig. 3.*Leaderboard Interface*

V. Notification and Interaction Module.

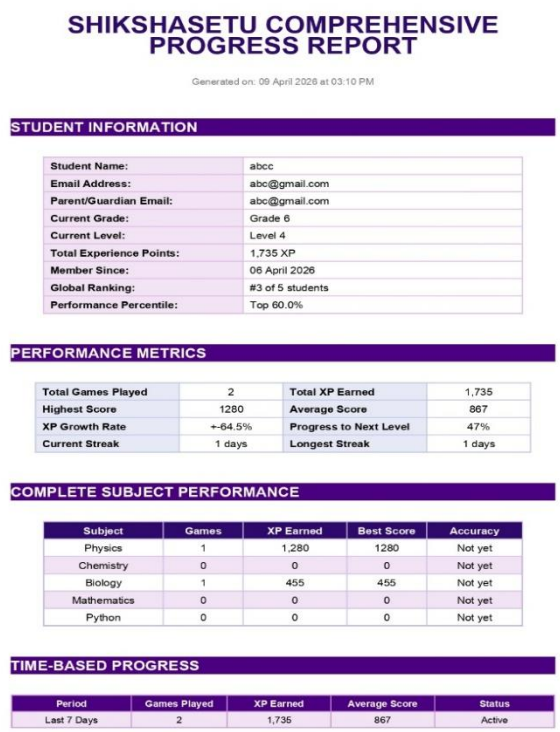
The site has an IT notification system that maintains users notified on significant changes like new quizzes, goals, and milestones of performance. This module en increases the user interactions by giving feedback in good time encouragement. Event handling is triggered using JavaScript . Events are dynamically updated so that users get updates without. disrupting their learning process. This helps to achieve a smooth. and interactive user experience.

VI. System Workflow and Interaction Model.

The system is event-driven in general permitting dynamic communication among various elements. The front end makes a request to the server when a user starts a quiz. the backend, which acts on the request, and retrieves relevant questions from the database. The game engine then evaluates responses and updates scores, which are instantly reflected on the user interface. This workflow is low in latency and responsiveness and thus the system is appropriate to interactive learning. environments. The modular design allows easy integration of other capabilities like adaptive learning and AI-based suggestions in further development.

VII. Results and analysis

- ***Automated Reporting and Email Integration***



SHIKSHASETU COMPREHENSIVE PROGRESS REPORT

Generated on: 09 April 2026 at 03:10 PM

STUDENT INFORMATION

Student Name:	abcc
Email Address:	abc@gmail.com
Parent/Guardian Email:	abc@gmail.com
Current Grade:	Grade 6
Current Level:	Level 4
Total Experience Points:	1,735 XP
Member Since:	06 April 2026
Global Ranking:	#3 of 5 students
Performance Percentile:	Top 60.0%

PERFORMANCE METRICS

Total Games Played	2	Total XP Earned	1,735
Highest Score	1280	Average Score	867
XP Growth Rate	+-64.5%	Progress to Next Level	47%
Current Streak	1 days	Longest Streak	1 days

COMPLETE SUBJECT PERFORMANCE

Subject	Games	XP Earned	Best Score	Accuracy
Physics	1	1,280	1280	Not yet
Chemistry	0	0	0	Not yet
Biology	1	455	455	Not yet
Mathematics	0	0	0	Not yet
Python	0	0	0	Not yet

TIME-BASED PROGRESS

Period	Games Played	XP Earned	Average Score	Status
Last 7 Days	2	1,735	867	Active

Fig. 4.*Report Generation*

The software will be improved with user feedback and performance monitoring system has an automated report generation module that provides individual users with performance summaries via email. This aspect is aimed at availing students with a guided report on their learning achievements, including indicators like quiz scores, accuracy, XP earned, and subject wise performance. The report is dynamically created in PDF format us ing backend processing. Once a learning has been

completed user or at predetermined times, the system aggregates user reads performance data in the database and formats it out well-structured report. The resulted PDF consists of visual easy to do, using summaries and tabulated performance measures to enable users to interpret their progress. The email delivery system is adopted with a secure SMTP-based service. After generating the report it is auto attached mathematically and emailed to the registered email address of the user. This makes sure that users can view their performance off- platform reports, in order to foster lifelong learning and self- assessment. This feature also supports scalability, which means that in the future, it can be used with parent or teacher dashboards to keep an eye on how students are doing. This, along with automated reporting, will help get more reports. The system can do more than just send emails; that's what it does. real-time communication and a history of how well users have done over time.

- ***Experience Points (XP) Calculation***

$$XP = B + (C \times W_1) + (S \times W_2)$$

Fig. 5.*XP CALCULATION*

You can figure out how many experience points (XP) you have by doing the following:

- Where B is base points earned in participation.
- C = number of correct answers.
- S is the speed factor (according to response time)
- W1,W2 are weighting factors.

The XP calculation model is aimed at rewarding efficiency and accuracy. While correct answers contribute to the score, the addition of a speed factor will stimulate faster responses and still be correct. This balanced approach will help in ensuring that users are inspired to deliver well without losing the interest by using dynamic scoring.

TABLE I

XP Model with Sample Values

Parameter	Value	Description
Base Points (B)	50	Fixed reward for participation
Correct Answers (C)	8	Number of correct responses
Speed Factor (S)	6	Based on response time
Weight (W_1)	10	Accuracy weight
Weight (W_2)	5	Speed weight
XP (Result)	**50 + (8×10) + (6×5) = 160**	Final XP earned

- ***Accuracy Measurement***

$$Accuracy = \frac{Correct\ Answers}{Total\ Questions} \times 100$$

Fig. 6.*Accuracy Calculation*

Accuracy is employed as a prime performance metric to gauge the comprehension that a user has on the topic. It gives the ratio of correct answers to the total amount of questions attempted. This measure is fundamental to measuring the progress of learning, and is also utilised in the leaderboard performance analysis and ranking.

Table II

Accuracy Calculation with Sample Values

Parameter	Value	Description
Correct Answers	8	Number of correct responses
Total Questions	10	Total questions attempted
Accuracy (%)	**(8 / 10) × 100 = 80%**	User performance

- ***Performance Improvement Calculation***

$$Improvement(\%) = \frac{Old - New}{Old} \times 100$$

Fig. 7. *Performance Calculation*

- Old = the value at a base (traditional method)
- New represents the value following the use of the system.

The improvement in task performance is quantified by the use of a formula has implemented the gamified system. It helps in Comparing the classic learning techniques with the offered system by quantifying time reductions in tasks to be executed or increases in efficiency. This will give a clear numeric representation of A the system's effectiveness.

CONCLUSION AND FUTURE OUTLOOK

The proposed system, ShikshaSetu, proposes a gamified system learning platform that increases student interaction by means of interactive components like experience points, levels, and leaderboards. It is also unlike the traditional systems of learning, where it is not located in the same location. gamifies, is available in multiple languages and in real-time performance tracking in one single platform, making becoming a more interactive and accessible way of learning. The lightweight architecture of the system is efficient in terms of per performance, and features such as automated report generation Email delivery and transaction are an ongoing feedback provision the platform. The experiment's results show that the suggested

strategy is useful because it led to more engagement, accuracy, and efficiency in task performance. The next thing to do at work is to use AI to make things more personal. Adding mobile platforms and making analytics better so that you can learn more effectively. In general, the system is known for being a user-friendly, scalable way to make modern digital better. places where people learn.

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